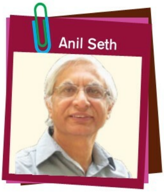


# Exploring Big Data on a Desktop, with HBase and Elasticsearch

The last couple of articles in this series on Big Data dealt with Elasticsearch. Extending this course of study further, let's see how Elasticsearch can be used to retrieve any document stored in HBase.

HBase (*http://hbase.apache.org/*) is an open source, non-relational database that runs on top of HDFS. It is modelled after Google's BigTable.

Consider the problem of storing a very large number of documents in their original format, whether in OpenOffice, PDF or MS Office, and then searching through them while trying to access any document randomly.

A key-value pair provides a convenient way to store the documents, where the key can be the file's name and the value of the contents of the document. You can easily use HBase as the data store for this purpose.

Elasticsearch has an attachment plugin *(https://github.com/elasticsearch/elasticsearch-mapper-attachments)* that uses Tika *(http://tika.apache.org/)* to analyse and index the attachments.

Hence, a combination of HBase and Elasticsearch (with the attachment type) can help you create a flexible solution that easily searches and selects from a huge number and variety of documents.

## Getting started with HBase

Currently, HBase is not available from the Fedora repository, though it will be for Fedora21. So, download the current version for an Apache mirror and just follow the quick start instructions (*http://hbase.apache.org/book/quickstart.html*). You could set it up in the fully distributed mode, using the three OpenStack virtual machines, *h-mstr, h-slv1* and *h-slv2*, as we have been doing in this series.

HBase is a Java application but comes with two servers, REST and Thrift, to allow integration with other programming environments, particularly Python. HappyBase (*http://happybase.readthedocs.org*) makes it easy to interface with Apache HBase from a remote machine; so, install it on the desktop.

Make sure that HDFS servers are running. Sign into *fedora@h-mstr*, start the HBase servers and create the 'Documents' table with one column, 'Content':

```
[fedora@h-mstr ~]$ cd hbase-0.96.2-hadoop2
[fedora@h-mstr hbase-0.96.2-hadoop2]$ bin/start-hbase.sh
[fedora@h-mstr hbase-0.96.2-hadoop2]$ bin/hbase-daemons.sh
start thrift
[fedora@h-mstr hbase-0.96.2-hadoop2]$ bin/hbase shell
hbase(main):001:0> create 'documents','content'
hbase(main):002:0> exit
```

The following Python code, *store_files_in_hbase.py*, scans all the files in a directory, selecting the ones you want. It then copies the file's contents in a variable, and stores the file name as the row key and the content as the column content in the 'Documents' table.

```
#!/usr/bin/python
import sys
import os
import happybase
TABLE='documents'
FILETYPES=['odt','doc','sxw','abw','pdf']
  def process_file(table,p,f):
    filename = '/'.join([p,f])
    filecontent = open(filename,'rb').read()
    print("Processing %s %s Len %d"%(p,f,len(filecontent)))
    table.put(filename,{'content:data':filecontent})
 def get_documents(path):
    for curr_path,dirs,files in os.walk(path):
        for f in files:
            try:
                if f.rsplit('.',1)[1].lower() in FILETYPES:
                    yield curr_path,f
            except:
                pass
try:
    path=sys.argv[1]
except IndexError:
    path='.'
connection = happybase.Connection('h-mstr')
table = connection.table(TABLE)
for p,f in get_documents(path):
    process_file(table,p,f)
```

It is best to index the documents at the same time as they are being stored in HBase. (As usual, in order to keep things simple, I have refrained from covering exception handling.)

## Elasticsearch attachments

Install the Elasticsearch attachment plugin on each of the three virtual machines and start the Elasticsearch server. For example, log in as *fedora@h-mstr:*